

PrimeFocus

November 2025



WHEN

November 21, 2025

Doors open at 7:00pm

Meeting starts at 7:30pm

Lecture starts at 8:00pm

WHERE

Unitarian Church

1893 North Vasco Road

Livermore, CA 94557

and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

Tri-Valley Star Gazers Astrophotography Imaging Group Panel

Mark Hai Du

Gert Gottschalk

Curtis Macchioni

Swaroop Shere

This month's meeting will be a panel presentation on astrophotography. The panel will be comprised of Gert Gottschalk, Mark Du, Curtis Macchioni, and Swaroop Shere. The panel will discuss the various aspects of astrophotography, including the equipment used, the techniques employed, and the results achieved. The panel will also discuss the various areas of astrophotography, including introductory astrophotography, planetary astrophotography, EAA and deep sky astrophotography. The presentations will be followed by a Q&A session.

Mark Du is an astrophotographer who began imaging in 2020 and has been learning and refining his craft ever since. By day, he works as a software engineer, bringing an analytical approach to solving the technical challenges of deep sky imaging. He currently shoots with an ASI2600MC Pro camera, ASKAR 120APO telescope, and ZWO AM5 mount.

Gert Gottschalk has been interested in all things astronomy since the 1980. His interest are building optics, making telescopes, and all kinds of observation and imaging in astronomy. He is TVS member since 1998. Gert has a PhD in electrical engineering from the university in Berlin, Germany and has professionally worked in the silicon valley tech industry since 1997. Gert Gottschalk has been interested in all things astronomy since the 1980. His interest are building optics, making telescopes, and all kinds of observation and imaging in astronomy. He is TVS member since 1998. Gert has a PhD in electrical engineering from the university in Berlin, Germany and has professionally worked in the silicon valley tech industry since 1997.

Curtis Macchioni is a physicist who retired after 40 years in the tech industry. He has shared his interest in Electronically Assisted Astronomy through numerous presentations, a article in the March 2016 issue of Sky & Telescope and his YouTube channel.

Swaroop Shere is a software engineer by profession and an amateur astrophotographer when time and family allows for it. He started with visual astronomy in the early 2010s before venturing into basic astrophotography around 2015. Over the years, he learnt the ropes mostly from other fellow astrophotographers, club members, youtube and other online resources. Currently he images with a remote setup consisting of AP-130 GT on paramount MYT with a QHY-268-M camera.

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
November 21	November 24	November 8
December 19	December 22	December 6
January 3	January 16	January 19

Money Matters

As of the last Treasurer’s Report on 10/20/25, our club’s account balance is \$46,403.32, this includes \$11,105.46 in the H2O Rebuild fund.

TVS Welcomes New Members

TVS welcomes new members Venkat Vinakollu, Kyla Dickerson Santana, Natalia McCoy, Amir Traboly, and Aarav Ahuja. Please say hello and chat with them during our meetings.

2025 TVS Club Star Party Schedule

Save the dates for the 2025 Club Star Parties. Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

November 22: Club/public star party at Del Valle Arroyo Road Staging Area, Arroyo Road Staging Area., Livermore. Set-up at 4:45pm, Observing 5:30-8:00pm.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

No events currently scheduled for Tesla Vintners.

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

No events currently scheduled for H2O.

CALENDAR OF EVENTS



November 21, 22, 28,29, December 5, 6, 12,13, 19, 20, 7:30-10:30pm

What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot’s historic telescopes. Chabot’s three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center’s 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see <https://chabotspace.org/events/events-listing/> <https://eastbayastro.org/chabot-telescope-status/>

December 8, 7:30 PM

What	Welcome to the Vera C. Rubin Observatory
Who	California Academy of Sciences
Where	Morrison Planetarium; 55 Music Concourse Drive, San Francisco, CA 94118
Cost	Public: \$15; Members and seniors: \$12

Featuring Dr. Phil Marshall, Stanford/SLAC

A couple of decades in the making, the world’s newest and fastest astronomical facility is now online and gearing up to carry out its primary mission, the 10-year Legacy Survey of Space and Time (LSST). Join Rubin’s Deputy Director of Operations, Phil Marshall, for a virtual tour of the NSF-DOE Vera C. Rubin

Observatory's Summit Facility in Chile, and hear more about Rubin and its mission.

The LSST will be an unprecedented astronomical data set for studies of the deep and dynamic universe, and is designed to be for everyone: Rubin will make the data widely accessible to a diverse community of scientists, and use it to enable the public to explore the Universe to greater depth and in more detail than ever before. In fact, the LSST Camera, built at SLAC and now mounted on Rubin's Simonyi Survey Telescope, presents everyone with a Big Data problem: how to cope with more astronomical data than has been taken by all previous telescopes combined, being accumulated and presented at unprecedented speed? Part of the answer to this question is "AI" - but what will that mean, in practice? Come and think big in the Morrison Planetarium.



The night sky dazzles over Rubin Observatory in this shot from October 2024. The Milky Way sprawls overhead in the waning light of sunset. Venus shines brightly on the left, while Comet C/2023 A3 (Tsuchinshan-ATLAS) appears just above the observatory at center. Image credit: Hernan Stockebrand

Phil is currently Deputy Director of Operations for the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST), setting up the Rubin Observatory and its survey to successfully deliver the data products needed by, for example, the cosmology analysis planned by the LSST Dark Energy Science Collaboration (DESC). He helped form the LSST DESC at its inaugural meeting in 2012, and held leadership positions in it for 7 years until he moved to his current position at Rubin. Phil was LSST DESC Spokesperson in 2017-2019, during which time he led the implementation of the collaboration's operations plan. His long-running scientific interest is strong gravitational lenses, whose Einstein rings and time delays can be used to probe the accelerated expansion history of the Universe, and which can help us probe the nature of Dark Matter via the sub-galactic structure than perturbs the lensing effect. Analyzing tens of thousands of these systems from the LSST will take new approaches to lens detection and modeling: Phil and his group at KIPAC are investigating machine learning with deep neural networks as a way to carry out principled, multi-level scientific inference at LSST scale. Phil did his PhD on Bayesian Analysis of Clusters of Galaxies at the University of Cambridge, in which

he first got interested in the process of measuring astronomical objects, including things like Dark Matter halos which we may not be able to observe directly. He first moved to Stanford in 2003 as one of KIPAC's first wave of postdocs, and returned as Kavli Fellow in 2009 after three years as TABASGO Fellow at the University of California, Santa Barbara. Phil then spent three years in Oxford as a Royal Society University Research Fellow, before moving back to join the SLAC staff on a permanent basis in 2013.

For more information, see <https://www.calacademy.org/events/benjamin-dean-astronomy-lectures/welcome-to-the-vera-c-rubin-observatory>





M31, and M13, By Ernest Epley



M31 and The Horsehead & Flame Nebulae, By Craig Siders

WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

November 2025

24	Mon	Algol shines at minimum brightness between 7:55 and 9:55 PM
28	Fri	Moon is at first quarter
29	Sat	Looking southeast the Moon is just $4\frac{1}{2}^\circ$ upper left of Saturn

December 2025

3	Wed	Moon rises with the Pleiades and eventually covers the Pleiades as the night continues
4	Thu	Full Moon
7	Sun	Just before sunrise look for the Moon just $4\frac{1}{2}^\circ$ below Pollux and just 5° right of Jupiter
9	Tue	In the Evening look for the Moon just $\frac{1}{2}^\circ$ above Regulus. as the night goes on the gap between them decreases
11	Thu	Moon is at last quarter
13-14	Sat-Sun	Geminid meteor shower peaks
14	Mon	Algol shines at minimum brightness between 9:39 and 11:39 PM

OFFICERS AND VOLUNTEER POSITIONS

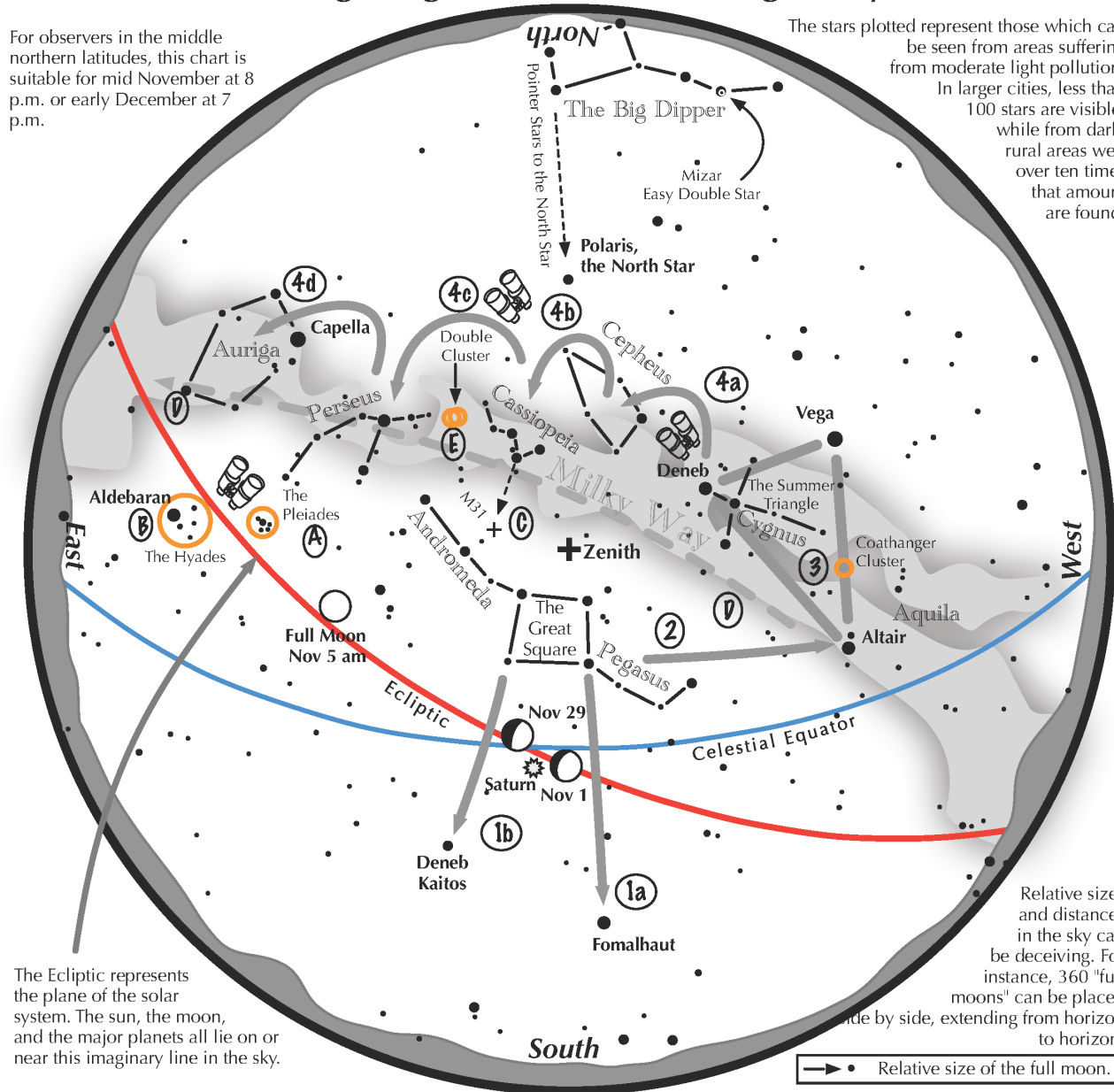
Officers	Volunteer Positions	Newsletter	Potluck Coordinator
President Eric Dueltgen president@trivalleystargazers.org	Astronomical League Rep. Don Dossa alrep@trivalleystargazers.org	Scott Schneider (Editor) newsletter@trivalleystargazers.org	OPEN potluck@trivalleystargazers.org
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Treasurer John Forrest treasurer@trivalleystargazers.org	Del Valle Coordinator David Wright delvalle@trivalleystargazers.org	Night Sky Network Rep. Ross Gaunt nnsn@trivalleystargazers.org	Publicity and Fundraising OPEN publicity@trivalleystargazers.org
Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H2O@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Johnathan Bailey outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

NAVIGATING THE NIGHT SKY FOR NOVEMBER 2025

Navigating the November Night Sky

For observers in the middle northern latitudes, this chart is suitable for mid November at 8 p.m. or early December at 7 p.m.

The stars plotted represent those which can be seen from areas suffering from moderate light pollution. In larger cities, less than 100 stars are visible, while from dark, rural areas well over ten times that amount are found.



The Ecliptic represents the plane of the solar system. The sun, the moon, and the major planets all lie on or near this imaginary line in the sky.

Relative sizes and distances in the sky can be deceiving. For instance, 360 "full moons" can be placed side by side, extending from horizon to horizon.

Navigating the November night sky: Simply start with what you know or with what you can easily find.

- 1 Face south. Almost overhead lies the "Great Square" with four stars about the same brightness as those of the Big Dipper. Extend a line southward following the Square's two westernmost stars. The line strikes Fomalhaut, the brightest star in the south. A line extending southward from the two easternmost stars, passes Deneb Kaitos, the second brightest star in the south.
- 2 Draw a line westward following the southern edge of the Square until it strikes Altair, part of the "Summer Triangle."
- 3 Locate Vega and Deneb, the other two stars of the Summer Triangle. Vega is its brightest member, while Deneb sits in the middle of the Milky Way.
- 4 Jump along the Milky Way from Deneb to Cepheus, which resembles the outline of a house. Continue jumping to the "W" of Cassiopeia, then to Perseus, and finally to Auriga with its bright star Capella.

Binocular Highlights

A and B: Examine the stars of the Pleiades and Hyades, two naked eye star clusters. **C:** The three westernmost stars of Cassiopeia's "W" point south to M31, the Andromeda Galaxy, a "fuzzy" oval. **D:** Sweep along the Milky Way from Altair, past Deneb, through Cepheus, Cassiopeia and Perseus, then to Auriga for many intriguing star clusters and nebulous areas. **E:** The Double Cluster.



Astronomical League www.astroleague.org/outreach; duplication is allowed and encouraged for all free distribution.



Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.